

CSE-251

ELECTRONIC CIRCUITS

LECTURE-04

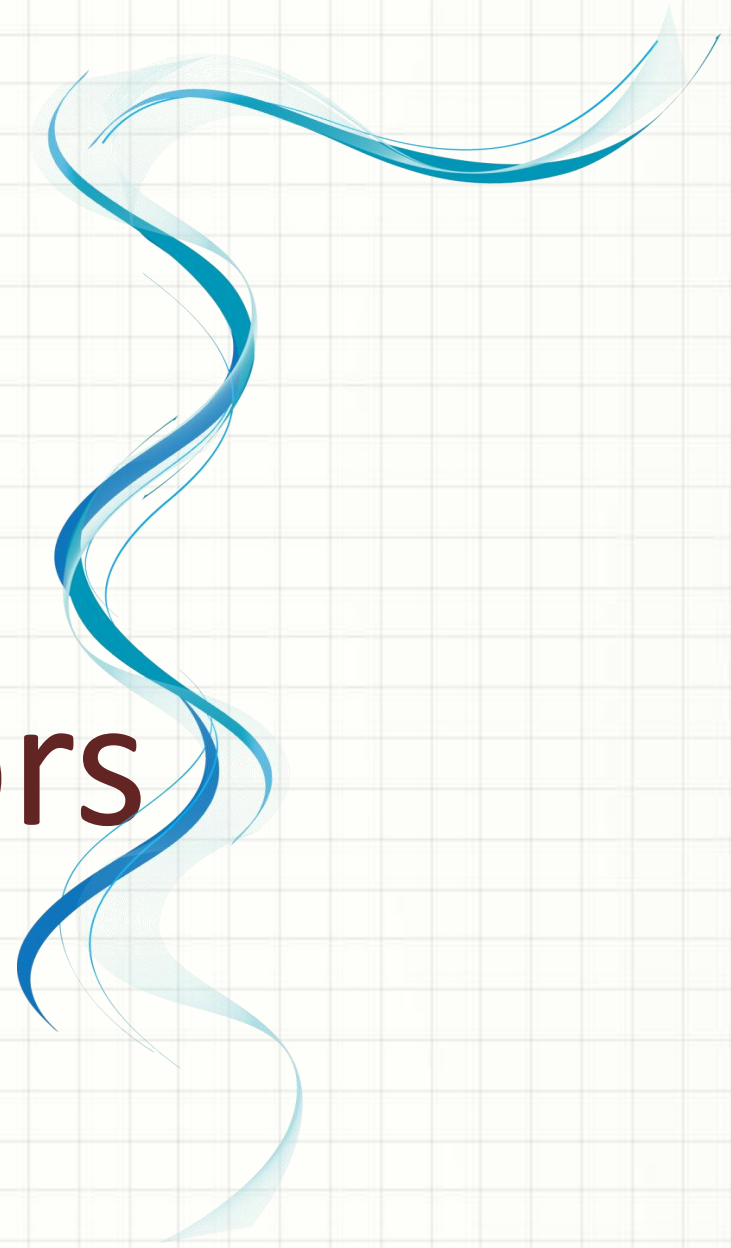
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Today's Overview:

- ❑ *All about Semiconductors*
- ❑ *Charge Carriers*
- ❑ *Applications*

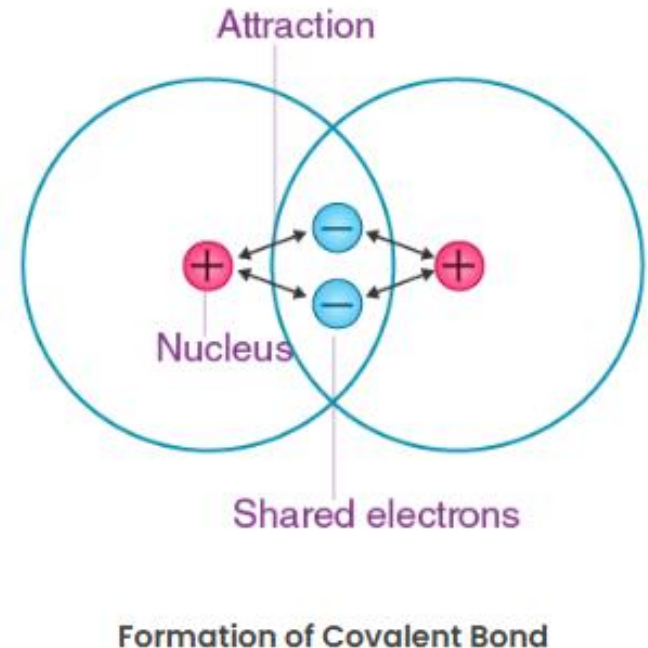


All about Semiconductors



Bonds in Semiconductors:

Elements having very high ionization energies are incapable of transferring electrons, and elements having very low electron affinity cannot take up electrons. The atoms of such elements tend to share their electrons with the atoms of other elements or with other atoms of the same element in such a way that both the atoms obtain octet configuration in their respective valence shells, and thus achieve stability. Such association through sharing of electron pairs among different or same kinds is known as **Covalent Bond**.



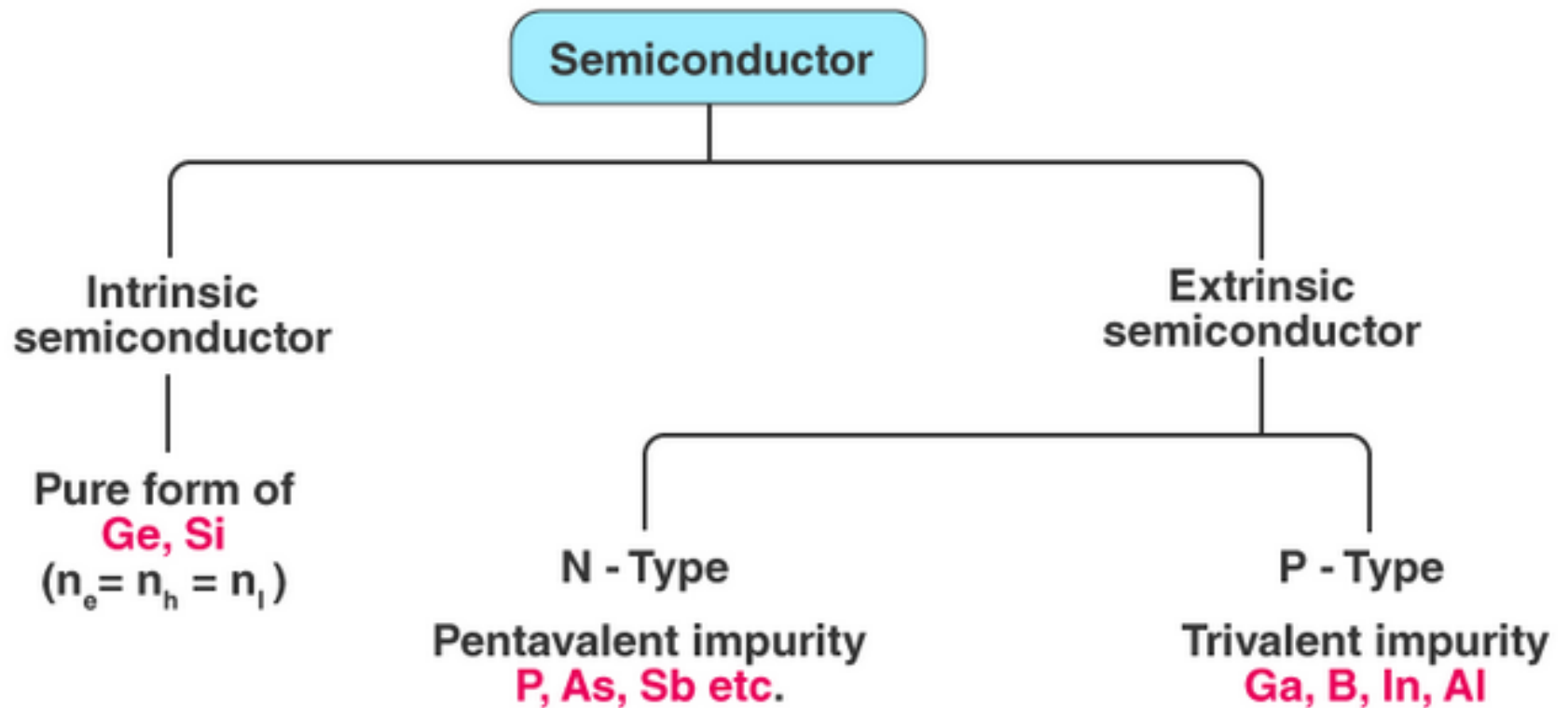
Nonmetals will readily form covalent bonds with other nonmetals in order to obtain stability, and can form anywhere between one to three covalent bonds with other nonmetals depending on how many valence electrons they possess.

A pair of electrons that is shared between two atoms is called a **bond pair**. A pair of electrons that is not shared between two atoms is called a **lone pair**.

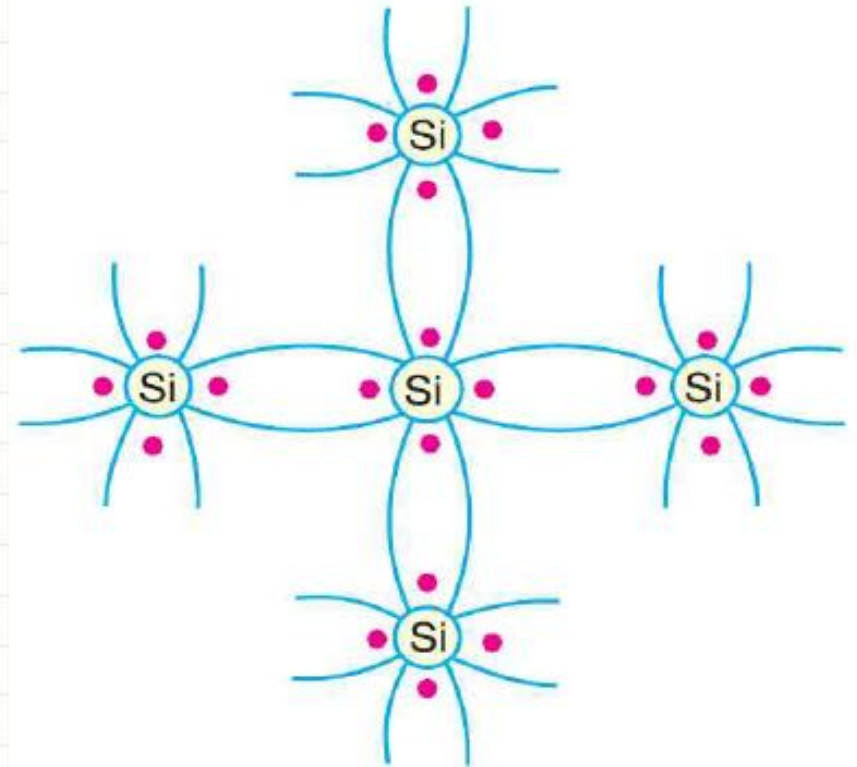
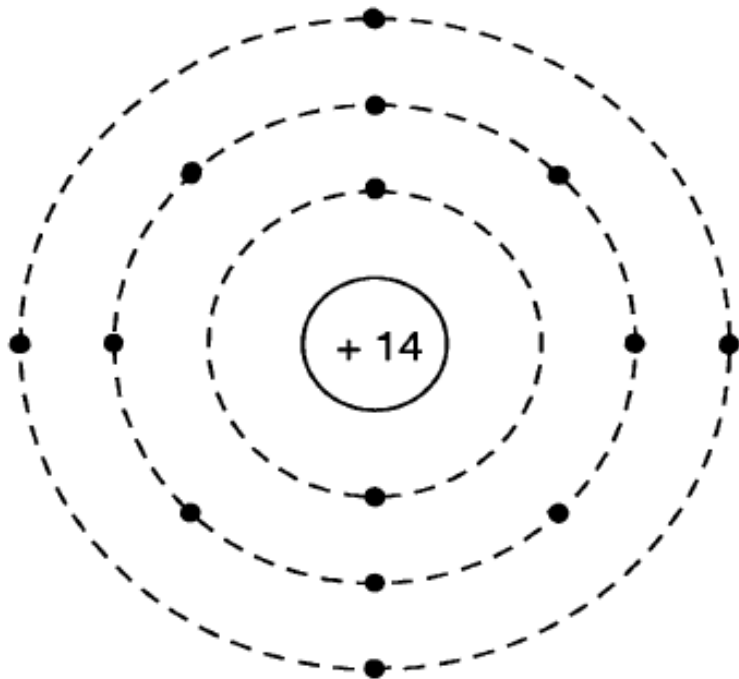
Properties of Semiconductors:

- ❑ *Semiconductor acts like an insulator at Zero Kelvin. On increasing the temperature, it works as a conductor.*
- ❑ *Due to their exceptional electrical properties, semiconductors can be modified by doping to make semiconductor devices suitable for energy conversion, switches and amplifiers.*
- ❑ *Lesser power losses.*
- ❑ *Semiconductors are smaller in size and possess less weight.*
- ❑ *Their resistivity is higher than conductors but lesser than insulators.*
- ❑ *The resistance of semiconductor materials decreases with the increase in temperature and vice-versa.*

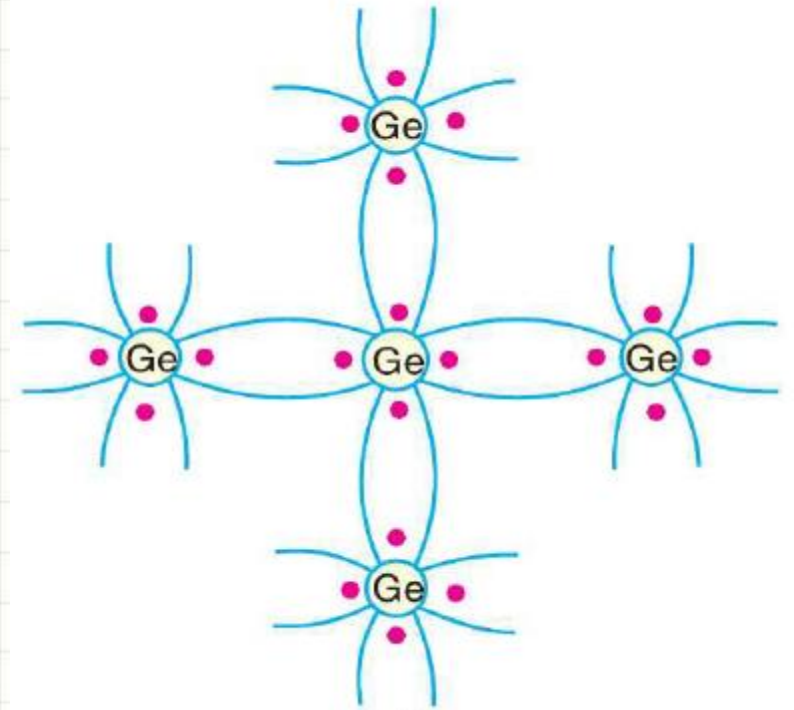
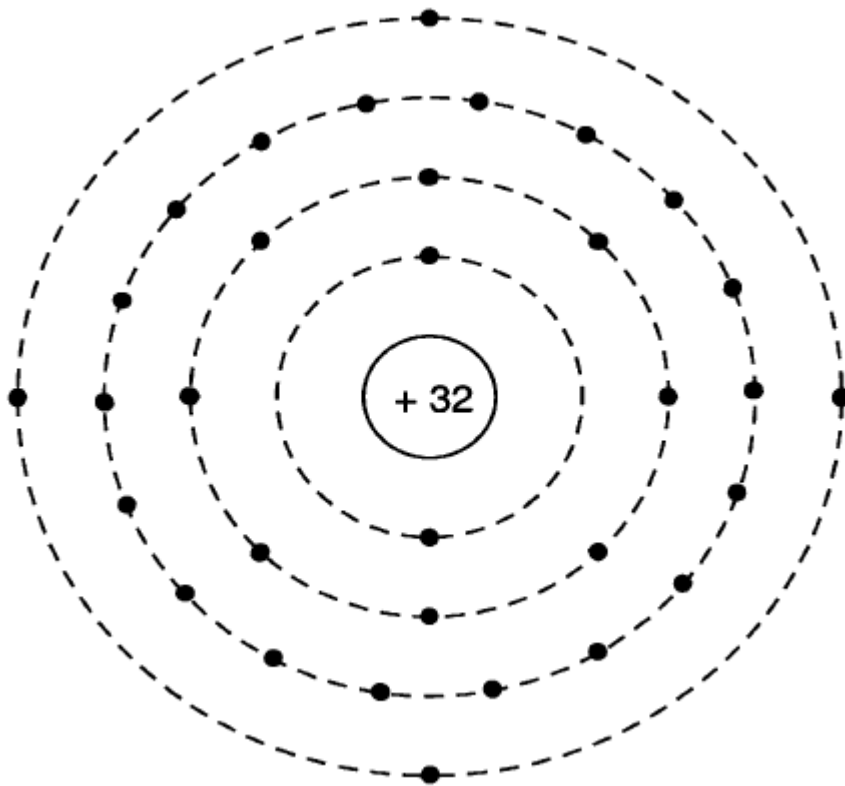
Types of Semiconductors:



Silicon structure:

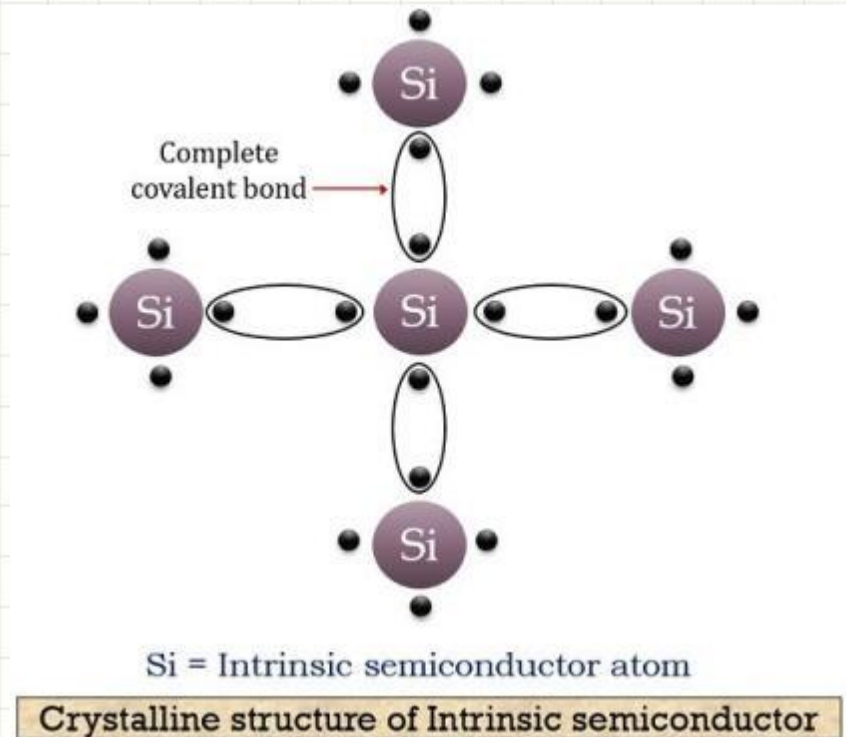


Germanium structure:



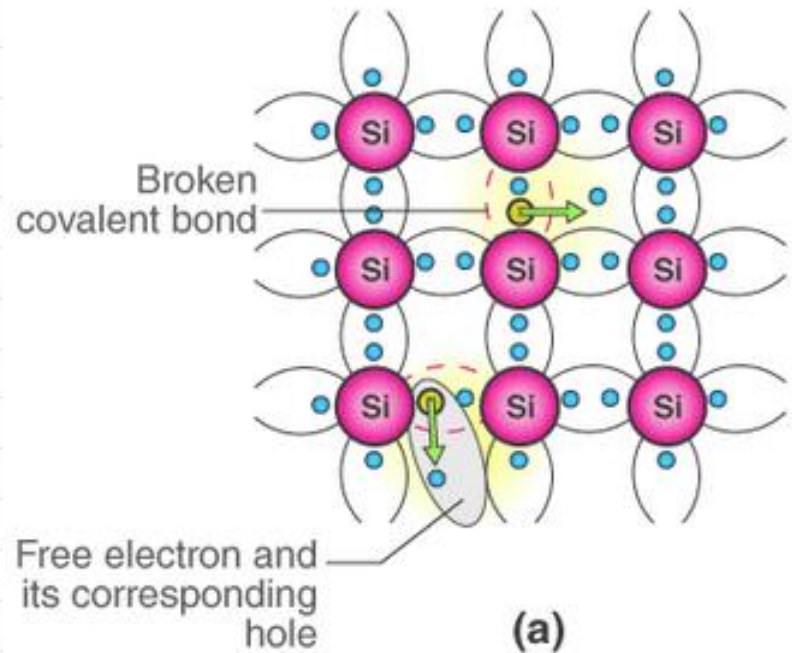
Intrinsic semiconductor:

- An **intrinsic type of semiconductor material** is made to be very pure chemically. It is made up of only a single type of element.
- Germanium (Ge) and Silicon (Si) are the most common type of intrinsic semiconductor elements. They have four valence electrons (tetravalent). They are bound to the atom by covalent bond.



Intrinsic semiconductor:

- When the temperature rises, due to collisions, few electrons are unbounded and become free to move through the lattice, thus creating an absence in its original position (hole). These free electrons and holes contribute to the conduction of electricity in the semiconductor. The negative and positive charge carriers are equal in number.



Extrinsic semiconductor:

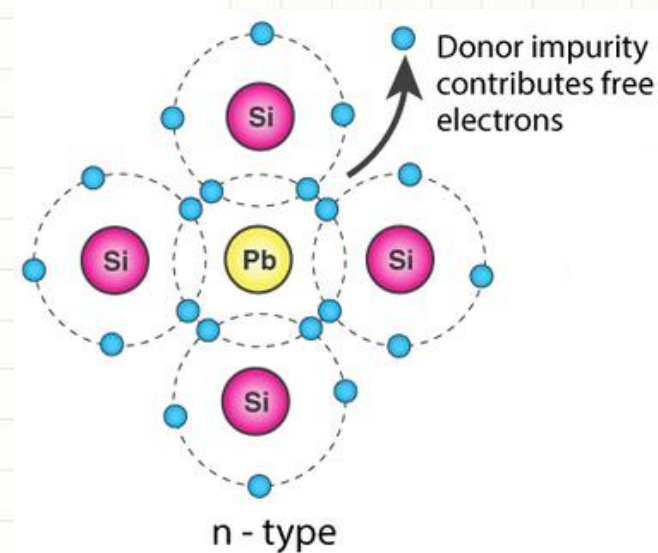
The conductivity of semiconductors can be greatly improved by introducing a small number of suitable replacement atoms called ***IMPURITIES***.

The process of adding impurity atoms to the pure semiconductor is called ***DOPING***.

Usually, only 1 atom in 10^7 is replaced by a dopant atom in the doped semiconductor.

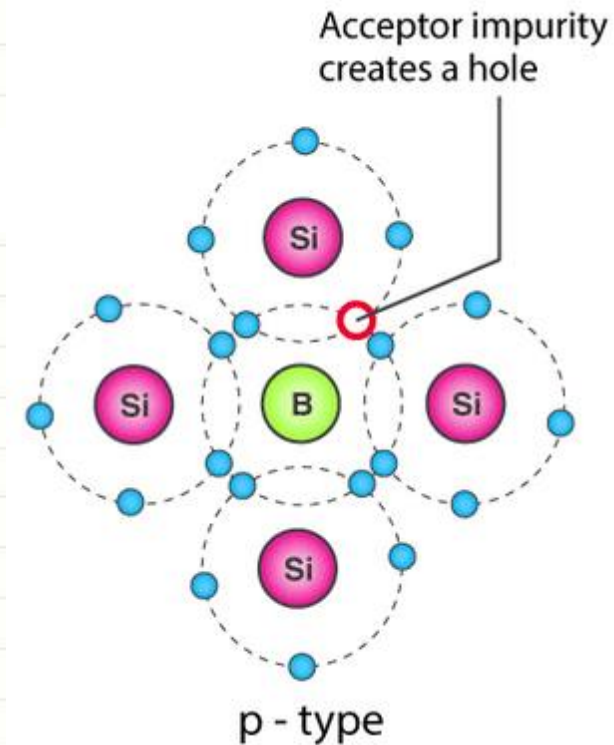
(Negative) n-type semiconductor:

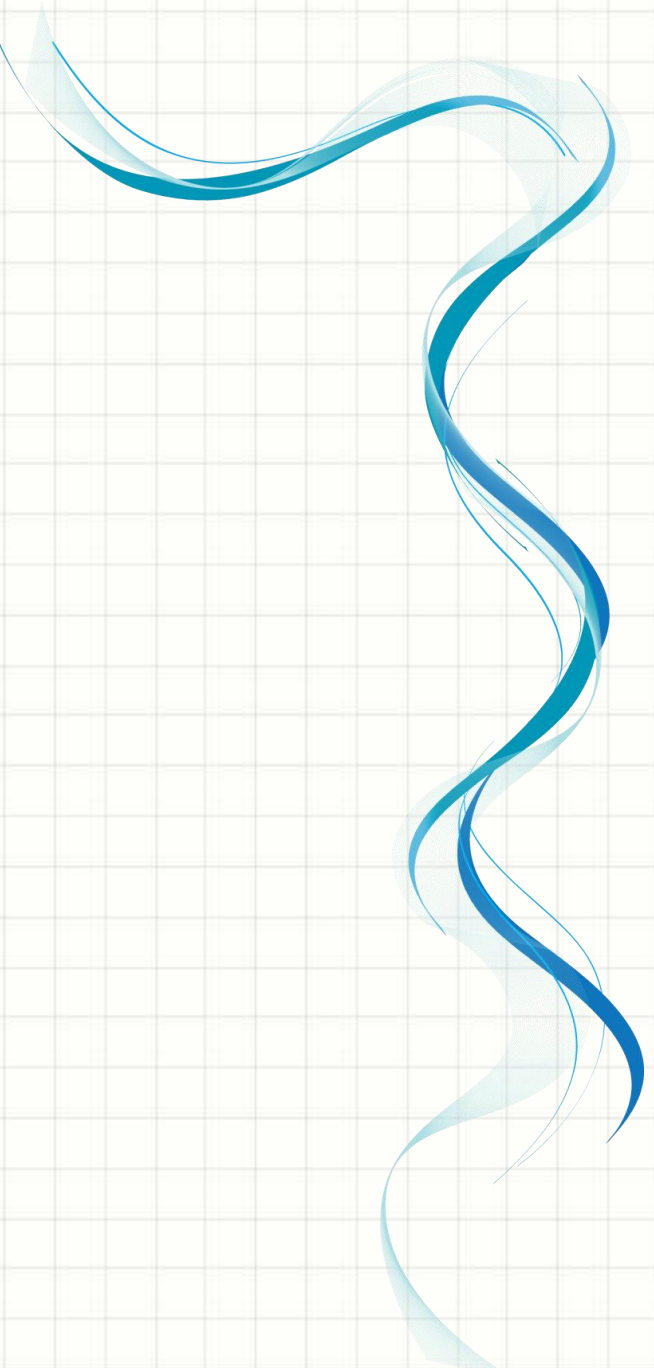
- ✓ When a pure semiconductor (Silicon or Germanium) is doped by **pentavalent impurity** (P, As, Sb, Bi) then, four electrons out of five valence electrons bonds with the four electrons of Ge or Si.
- ✓ The fifth electron of the dopant is set free. Thus, the impurity atom donates a free electron for conduction in the lattice and is called “**Donar**”.
- ✓ Since the number of free electron increases by the addition of an impurity, the negative charge carriers increase. Hence, it is called ***n-type semiconductor***.
- ✓ As conduction is due to a large number of free electrons, the electrons in the n-type semiconductor are the **MAJORITY CARRIERS** and holes are the **MINORITY CARRIERS**.



(Positive) p-type semiconductor:

- ✓ When a pure semiconductor is doped with a **trivalent impurity** (B, Al, In, Ga), then the three valence electrons of the impurity bonds with three of the four valence electrons of the semiconductor.
- ✓ This leaves an absence of electron (hole) in the impurity. These impurity atoms which are ready to accept bonded electrons are called “**Acceptors**”.
- ✓ With the increase in the number of impurities, holes (the positive charge carriers) are increased. Hence, it is called ***p-type semiconductor***.
- ✓ The holes in the p-type semiconductor are **MAJORITY CARRIERS** and electrons are **MINORITY CARRIERS**.



A decorative blue wavy line on the left side of the slide, consisting of multiple overlapping, flowing curves in various shades of blue.

Charge Carriers

Charge Carriers:

Generally, ***Carrier*** refers to any object that carry another object from one place to another place. For example, in countries such as India, Singapore and Brazil: Tiffin box or Tiffin carriers are widely used for carrying food from one place to another place. Here, the Tiffin box acts as a carrier that carries the food from one place to another place. Similarly, people use vehicles such as buses, trains, airplanes, etc. to travel from one place to another place. Here, the vehicles act as carriers that carry people from one place to another place.

In the similar way, ***particles such as free electrons and holes carry the charge or electric current from one place to another place. These are known as “Charge carriers”.***

Types of Carriers:

There are two basic types of charge carriers:

- Positive charge carriers and
- Negative charge carriers

➤ Positive charge carriers:

The positive charge carriers (such as **holes**) are the carriers that carry positive charge with them while moving from one place to another place. Holes are the vacancies in valence band that moves from one place to another place within the valence band.

➤ Negative charge carriers:

The negative charge carriers (such as **free electrons**) are the carriers that carry negative charge with them while moving from one place to another place. Free electrons are the electrons that are detached from the parent atom and moves freely from one place to another place.

Types of Carriers:

Based on the quantity of carriers, charge carriers can also be divided into two types:

- Majority charge carriers and
- Minority charge carriers

➤ Majority charge carriers:

The charge carriers that are present in large quantity are called “**Majority charge carriers**”. The majority charge carriers carry most of the electric charge or electric current in the semiconductor. Hence, majority charge carriers are mainly responsible for electric current flow in the semiconductor.

➤ Minority charge carriers:

The charge carriers that are present in small quantity are called “**Minority charge carriers**”. The minority charge carriers carry very small amount of electric charge or electric current in the semiconductor.

Charge carriers in Semiconductors:

□ Intrinsic semiconductor:

The semiconductors that are in pure form are called intrinsic semiconductors. In intrinsic semiconductor, the total number of negative charge carriers (free electrons) is equal to the total number of positive charge carriers (holes).

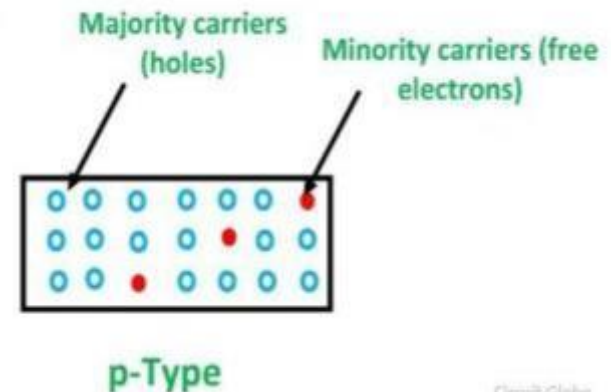
Total negative charge carriers = Total positive charge carriers

Charge carriers in Semiconductors:

□ P-type semiconductor:

In p-type semiconductor, large number of holes is present. Hence, holes are the **majority charge carriers** in the p-type semiconductor. The holes carry most of the electric charge or electric current in the p-type semiconductor.

In p-type semiconductor, very small number of free electrons is present. Hence, free electrons are the **minority charge carriers** in the p-type semiconductor.



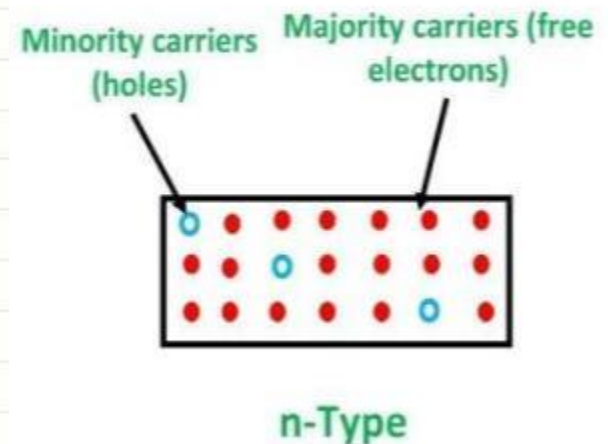
Total negative charge carriers < Total positive charge carriers

Charge carriers in Semiconductors:

□ N-type semiconductor:

In n-type semiconductor, large number of free electrons is present. Hence, free electrons are the **majority charge carriers** in the n-type semiconductor. The free electrons (majority charge carriers) carry most of the electric charge or electric current in the n-type semiconductor.

In n-type semiconductor, very small number of holes is present. Hence, holes are the **minority charge carriers** in the n-type semiconductor.



Total negative charge carriers > Total positive charge carriers

Applications of Semiconductors:

- Semiconductors are used in almost all electronic devices. Without them, our life would be much different.
- Their reliability, compactness, low cost and controlled conduction of electricity make them ideal to be used for various purposes in a wide range of components and devices.
- Transistors, diodes, photo-sensors, microcontrollers, integrated chips (ICs) and much more are made up of semiconductors.

Uses in Daily Life:

- Temperature sensors are made with semiconductor devices.
- They are used in 3D printing machines
- Used in microchips and self-driving cars
- Used in calculators, solar plates, computers and other electronic devices.
- Transistor and MOSFET used as a switch in Electrical Circuits are manufactured using the semiconductors.

Uses in Industry:

- The physical and chemical properties of semiconductors make them capable of designing technological wonders like microchips, transistors, LEDs, solar cells, etc.
- The microprocessor used for controlling the operation of space vehicles, trains, robots, etc is made up of transistors and other controlling devices which are manufactured by semiconductor materials.



QUESTIONS?

SEE YOU IN THE NEXT CLASS

